Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

Name: Monish P

Roll No: 250701446

EXPERIMENT 6

Time series analysis of website traffic

Experiment 6: Time series analysis of website traffic

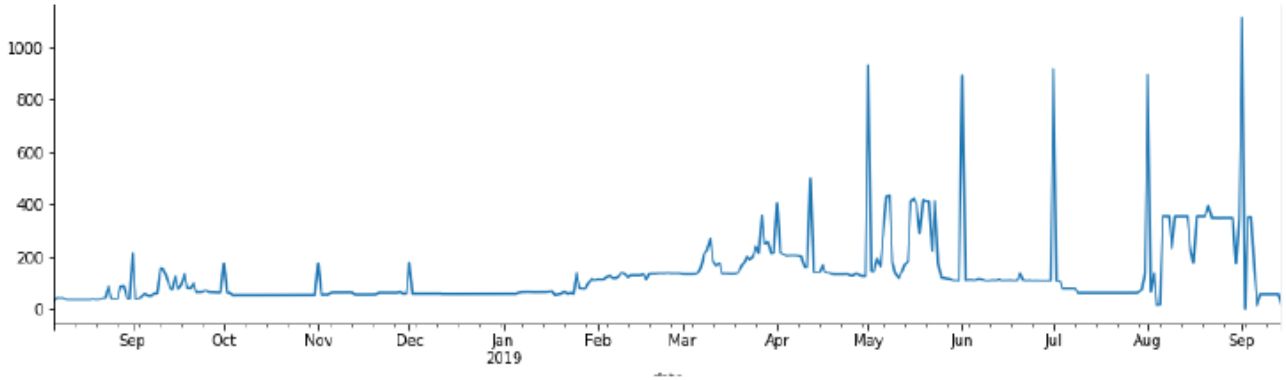
Aim

To analyze website traffic over time and identify trends, seasonality, and anomalies.

Source Code

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')
decomposition = seasonal_decompose(data['Page_Views'], model='additive')
decomposition.plot()
plt.show()
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()
data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /
data['Page_Views'].std()
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')
plt.legend()
plt.show()
```

Output - Experiment 6



Output - Experiment 6

Expected Outcomes:

 Clear visualization of trends, seasonality, and anomalies in website traffic.

 Understanding of the underlying patterns in user behavior over time.

24

Result - Experiment 6

Website traffic time-series data was analyzed for trends, seasonality and anomalies.